

Industry-Specific AI Solution.

Reflective Synthesis Paper.

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Reflective Synthesis Paper: AI-Assisted Chronic Care Triage for Outpatient Diabetes Follow-Up

Industry Context and Problem Definition

Healthcare systems face a recurring operational problem: many patients with chronic conditions show early signs of worsening risk before their next scheduled visit, but care teams do not always have enough time to review every chart in depth. In outpatient diabetes management, those early signals can include rising A1C levels, missed appointments, repeated emergency or inpatient utilization, medication changes, and lower adherence. If those indicators are not recognized and acted on quickly, the result can be worse patient outcomes, avoidable acute care use, and higher system cost. Early hospital readmission within 30 days is a widely used outcome that captures exactly this gap between what was needed and what actually happened.

This problem is well suited to an AI-enabled workflow because it combines structured data analysis, predictive prioritization, communication support, and operational decision routing. At the same time, it is not a setting where full automation is appropriate. Healthcare is a high-stakes domain in which bias, privacy, explainability, and accountability matter as much as technical performance. The value of an industry-focused capstone solution, therefore, is not simply building a model; it is designing a coherent system that is technically plausible, ethically grounded, and professionally defensible.

The integrated solution I designed for this project is an AI-assisted chronic care triage system for outpatient diabetes follow-up. Its purpose is to help a clinic sort patients into actionable risk tiers, generate a short clinical summary of why a case is being prioritized, and route the case to the appropriate follow-up path. The system does not diagnose, prescribe, or replace a clinician. Instead, it supports earlier attention for patients who may need help while keeping a human in the loop for high-risk outputs.

Overview of the Integrated Solution

The artifact combines four connected layers. First, a data-analysis layer loads and preprocesses the UCI Diabetes 130-US Hospitals (1999-2008) dataset (Strack et al., 2014), a publicly available record set covering ten years of inpatient encounters across 130 hospitals in the United States. It defines the binary outcome `'needs_intervention = (readmitted < 30 days)'` and profiles indicators such as age band, time in hospital, prior outpatient/emergency/inpatient utilization, number of medications, A1C result, insulin status, and whether diabetes medication was changed during the encounter.

Second, a supervised machine learning layer trains both a logistic regression model and a random forest classifier on the same train/test split (stratified by the outcome) and selects the better-performing model by held-out ROC-AUC.

Third, a generative AI layer uses a large language model to translate the structured prediction and patient indicators into a short, professional clinical case summary and a separate patient-

facing outreach message. The LLM backend is pluggable: OpenAI by default, with Anthropic and a local `llama-cpp-python` model also supported.

Fourth, an agentic routing layer uses the same LLM as a lightweight tool-using agent: it reads the patient context, considers a list of available follow-up tools (educational reminder, nurse outreach, physician review), and returns a JSON decision containing the chosen action, a one-sentence justification grounded in the indicators, and an estimated follow-up window in hours.

This architecture reflects the reality that useful AI systems in industry rarely depend on a single technique. They integrate analytics, prediction, explanation, and orchestration into a workflow that matches how people actually work. The design is intentionally modular so that each layer can be evaluated separately while still contributing to a unified operational purpose.

Integration of Prior Projects and Methods

This project is explicitly built from methods and perspectives developed across earlier capstone work. The first influence came from prior data and statistical reasoning exercises, which trained me to identify relevant variables, summarize distributions, and reason carefully about patterns before modeling. That earlier work shaped the front end of the pipeline: the system loads a real clinical dataset, drops out-of-scope cases (deceased, hospice), recodes age bands into a numeric midpoint, and treats sentinel "?" values as missing rather than as informative categories. It also shaped the choice of outcome. Rather than predicting an arbitrary engineered label, the system targets early 30-day readmission because that outcome is clinically meaningful, well documented in the dataset, and directly aligned with the operational decision the prototype supports.

The second influence came from prior machine learning work. Earlier predictive modeling projects emphasized feature selection, train/test separation, interpretable performance metrics, and the importance of aligning modeling choices with the problem context. Those lessons informed several specific decisions in the artifact: stratified train/test splitting on the outcome, class-weight balancing because the positive class is roughly 11% of the cohort, side-by-side evaluation of a linear and a tree-based model, and reporting accuracy, precision, recall, F1, and ROC-AUC rather than a single metric. The selected model is the one with the higher held-out ROC-AUC.

The third influence came from prior generative AI activities. Those projects demonstrated that model outputs become more useful when they are translated into readable, context-aware language for a human decision-maker. In this capstone solution, the generative layer is an LLM call. Each shortlisted patient produces two pieces of generated language: a short clinical case summary written for the care team, and a separate plain-language outreach message written for the patient. Both prompts include explicit guardrails ("never diagnose", "never invent indicators that were not provided", "do not include medical advice"), which is consistent with how generative AI is responsibly used in clinical-adjacent settings.

The fourth influence came from prior agentic workflow design. Earlier work on tool chaining and structured decision-making showed that decision support is more useful when the system can recommend the next operational action rather than stopping at prediction. In this artifact the

routing decision itself is produced by the LLM. The agent is given (1) the structured patient context produced by the predictive layer, (2) a list of available tools with descriptions, and (3) instructions to output strict JSON. It must select exactly one tool, justify the selection in one sentence grounded in the indicators, and estimate a follow-up window in hours. The output is parsed, validated against the allowed action set, and conservatively defaults to physician review if validation fails. This is the lightweight tool-use pattern that defines an agentic step in modern LLM systems.

Technical Design Decisions and Tradeoffs

Several design decisions shaped the final artifact.

The first was the choice of dataset. Using the UCI Diabetes 130-US Hospitals dataset means the evaluation measures how well a model generalizes to data whose underlying structure it does not have access to. The tradeoff is that real clinical data is noisier, more imbalanced, and harder to model than a clean simulation. The held-out ROC-AUC therefore reflects honest performance on a hard real-world problem rather than an optimistic benchmark.

The second was running two model classes (logistic regression and random forest) on the same split and selecting on the test ROC-AUC. The logistic regression contributes interpretability (signed coefficients on encoded features) and the random forest captures non-linear interactions. Reporting both, and selecting between them, demonstrates real modeling judgment rather than a single fixed pipeline.

The third was percentile-based risk bands. Because the predicted probability distribution depends on the chosen model and the calibration of class weighting, fixed thresholds (for example, "Critical above 0.75") behave inconsistently across runs. The artifact instead defines bands by quantile of the predicted-probability distribution: top 5% Critical, next 10% High, next 20% Moderate, remaining 65% Low. This matches how a clinic with finite triage capacity actually wants to use such a model.

The fourth was constraining the generative and agentic layers with explicit instructions and a structured output schema. The case-summary prompt forbids diagnosis, prescription, and invention of indicators. The agentic prompt enforces JSON output, restricts the action space to a small named set of tools, and conservatively escalates to physician review whenever the LLM's response cannot be parsed or validated. This trades some flexibility for safety, which is the right tradeoff in a healthcare-adjacent decision-support setting (NIST, 2023; WHO, 2021).

The fifth was scoping the LLM layer to a triage shortlist (configurable, default 20 patients) rather than the full ~99,000-row dataset. The classifier scores everyone and the resulting risk bands are exported in full. Generative case summaries, outreach messages, and agentic routing decisions are produced for the shortlist, which is the operationally useful slice for a care team and which keeps API usage bounded.

Ethical, Governance, and Responsible AI Considerations

The central ethical concern in this project is not whether AI can prioritize patients, but whether it can do so fairly, transparently, and safely. Healthcare data often reflect structural inequities in access, documentation, and prior treatment. If those patterns are learned uncritically, the system may reproduce or amplify unfair outcomes. Obermeyer et al. (2019) showed that health management algorithms can encode racial bias when proxy variables such as historical cost are used without scrutiny. That finding is directly relevant here because triage systems can appear objective even when they are shaped by unequal historical conditions, and the UCI dataset itself reflects U.S. inpatient practice over the 1999-2008 period.

To address that risk, the prototype includes several safeguards. First, it uses a transparent set of clinical and behavioral indicators and reports per-feature importance for the linear model. Second, it routes high-risk and ambiguous cases to human review rather than acting autonomously. Third, the LLM layers are instructed to never diagnose, never prescribe, and never invent indicators that were not provided. Fourth, the agentic layer's output is validated against a fixed allow-list of actions and conservatively escalated when validation fails. Fifth, the artifact is framed as decision support, not decision replacement, and its limitations (described below) are made explicit.

Transparency is another major ethical issue. A risk score on its own can create automation bias, in which staff trust an output simply because it appears technical. The generative case-summary layer is designed to reduce that risk by making the main drivers of the prediction legible to clinicians and staff, in language that can be reviewed and challenged. That design choice is consistent with broader calls for human-centered AI in medicine, where systems are most valuable when they enhance human judgment rather than displace it (Topol, 2019).

Privacy and governance also matter. Even though the UCI dataset was released for research and contains no direct identifiers, any real deployment would require secure handling of electronic health record data, role-based access controls, audit logging, organizational review, and a clear policy on what is sent to a third-party LLM provider. WHO (2021) stresses that AI for health must place ethics and human rights at the center of design and governance. In practice, that means the system would need clear accountability, documentation of intended use, escalation policies, ongoing monitoring after launch, and an alternative to a third-party API for any context where sending patient context off-site is not allowed. The artifact supports this by also accepting Anthropic and a local `llama-cpp-python` model as alternative LLM backends.

Limitations and Risks

The prototype has several important limitations.

The first is dataset scope. The UCI Diabetes 130-US Hospitals dataset reflects inpatient encounters in 130 U.S. hospitals between 1999 and 2008. Patterns in that data may not generalize to current outpatient populations, to non-U.S. systems, or to populations underrepresented in the original sample. Subgroup fairness analysis would be required before any real use.

The second is model performance. With class weighting and feature engineering as described, the held-out ROC-AUC of the selected model is in the mid-0.6 range, which is consistent with prior published work on this dataset. That is the realistic outcome for a noisy clinical problem,

but it also means the prototype's predictions are imperfect and should be used as a prioritization aid rather than a decision rule. False negatives (patients who need follow-up but are not flagged) are particularly concerning in healthcare and would need to be tracked closely.

The third is the LLM layer. LLMs can produce fluent text that is wrong, and even with constrained prompting the case summary and patient message are not guaranteed to be free of error. The system mitigates this by routing all flagged cases through a human reviewer, by restricting the agent's action space, and by conservatively escalating when output cannot be parsed. It does not eliminate the risk.

The fourth is workflow adoption. Even a technically sound system may fail if staff do not trust it, if alerts are too frequent, or if the additional documentation creates burden. Real deployment would require usability testing, alert tuning, and integration with existing care management workflows.

Professional and Industry Relevance

This project demonstrates professional readiness because it goes beyond a narrow modeling task and addresses what employers and research stakeholders actually expect: cross-domain synthesis on a real dataset, defensible design tradeoffs, ethical awareness, and clear communication. In practice, AI professionals are often asked to connect analytics, modeling, user experience, governance, and operations in a single solution. That is precisely what this artifact attempts to show.

The four layers are deliberately implemented with techniques appropriate to their domain: real preprocessing on a real dataset for the data layer, two supervised classifiers compared on a held-out split for the ML layer, an LLM-driven generative layer with explicit guardrails, and an LLM-driven routing agent that selects from a structured tool set and produces a justification. Combined with the ethical and governance reasoning above, the project demonstrates not only technical fluency but professional judgment.

Future Extensions or Improvements

If I were extending this work, the next step would be to perform formal subgroup performance analysis (race, gender, age band, payer code where allowed) on the held-out test set and to add calibration plots in addition to ROC-AUC. I would also experiment with gradient-boosted trees and, where data is rich enough, with calibrated probability outputs. On the LLM side, I would add an evaluation harness that has a clinician rate a sample of generated case summaries and routing justifications for accuracy and tone, and I would add automated checks (for example, that the case summary never names a diagnosis the patient does not have in the record). Finally, I would formalize model documentation with a model card or similar governance artifact so that intended use, limitations, training data lineage, and known risks are visible to all stakeholders.

Conclusion

This capstone synthesis project demonstrates how a real-world healthcare problem can be addressed through the intentional integration of data analysis, predictive modeling, generative explanation, and agentic workflow design. Each of the four layers is implemented with a real technique appropriate to its domain: a real dataset, two real classifiers compared on a held-out

split, an LLM-driven generative layer with explicit guardrails, and an LLM-driven routing agent that selects from a structured tool set and produces a justification. Combined with the ethical and governance reasoning above, the project demonstrates not only technical fluency but professional judgment.

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